



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2015 - 235



**ZOOMLION Rice Combine Harvester
Model: 4LZT-4.0ZD**

ZOOMLION 4LZT-4.0ZD RICE COMBINE HARVESTER

Test Report No.	:	2015 - 235
Date of Test	:	October 10, 2015
Test Valid Until	:	October 10, 2020
Brand	:	ZOOMLION
Model	:	4LZT-4.0ZD
Type	:	Ride-on, crawler-type, tank-type
Manufacturer	:	ZOOMLION HEAVY MACHINERY CO., LTD. Exi 16, Sanshan District, Wuhu, Anhui China
Test Requested By	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The ZOOMLION 4LZT-4.0ZD Rice Combine Harvester was tested using "Hybrid" rice variety with a Grain Moisture Content of 20.7%, wet basis, Grain-Straw Ratio of 0.41 and average Plant Height of 970 mm. Based on the results of the test, the ZOOMLION 4LZT-4.0ZD Rice Combine Harvester had an Actual Field Capacity of 0.378 ha/h or 2.65 h/ha and Field Efficiency of 76.1% at an average Travelling Speed of 4.25 kph. It had also a Threshed Grain Output Capacity of 4546 kg/h with threshed grain Purity of 97.31 %. The rice combine harvester had a Total Grain Loss of 0.14% consisting of 0.11% Header Loss, 0.03% Separation/Blower Loss, no Uncut Loss and no Unthreshed Loss.

Grain quality analysis of threshed grains showed a net percent Cracked Grain of 0.84 % and 0.02% Broken Grains.

The Fuel Consumption Rate of the rice combine's diesel engine was 6.42 L/h.

During the test, four persons were needed to operate the rice combine: one for operating the machine, two for bagging and one for tying the sack of paddy.

No breakdown or failure of components occurred during the test

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at a farmer's field in Nanling County, Wuhu City, China on October 10, 2015 and was evaluated on the following items:

1. Field Capacity
2. Field Efficiency
3. Output Capacity
4. Grain losses
 - a. Header Loss
 - b. Separation/Blower Loss
 - c. Uncut Loss
 - d. Unthreshed Loss
5. Fuel Consumption
6. Labor Requirement

OPERATOR'S MANUAL

The rice combine harvester was provided with Operation and Maintenance Manual which contains the following: Safety Instructions; Safety Signs and Positions; Operation Instructions; Travelling and Transportation; Maintenance and Servicing; Schedules; and, After-Sales Service and Warranty.

DESCRIPTION OF THE MACHINE

The ZOOMLION 4LZT-4.0ZD Rice Combine Harvester (**Figure 1**), was a ride-on, crawler-type, tank-type harvesting machine which combined the cutting and threshing operations in one machine. It was powered by a 65.7 kW diesel engine which drove the transmission systems of the machine. The rice combine consisted of reel assembly, cutting assembly, screw and chain conveyors, threshing assembly, traveling device and control levers.

The reel assembly (**Figure 2**) with its rotary motion guides the rice plant to the cutting assembly. It consisted of five anchor bars and each anchor bar consisted of 15 pieces of protruding prongs.

The machine's cutting device consisted of reciprocating cutting knives (**Figure 3**). It had a cutting width of 2.0 m and the height of cut was adjustable.

The rice combine's delivery device to its threshing unit consisted of a screw conveyor (**Figure 4a**) and a chain conveyor (**Figure 4b**). The screw conveyor conveyed the cut plants to a chain conveyor which in turn delivered the cut plants to the threshing chamber of the combine.

The threshing assembly was an axial-flow type and consisted of an open-type cylinder with peg teeth (**Figure 5**).

The cleaning assembly consisted of a centrifugal fan which separated the chaffs, straws, empty grains and other light impurities from the threshed grain and a concave screen.

The rice combine was provided with a grain tank (**Figure 6a**) which held the threshed grain and a folding-type discharge chute (**Figure 6b**) with screw conveyor where threshed grains were discharged.

The rice combine harvester was provided with control levers and control panel (**Figure 7**). The control levers were the following: main shift lever; auxiliary shift lever; throttle control lever; steering and header lift handle; reel lift handle; and, hydraulic grain unloading handle. The control panel houses the fuel meter, tachometer (for engine speed) and water temperature meter.

The machine had a crawler-type (**Figure 8**) traveling device, with a width of 500 mm.



Figure 1. ZOOMLION Rice Combine Harvester
Model: 4LZT-4.0ZD



Figure 2. Reel Assembly



Figure 3. Reciprocating cutting knife

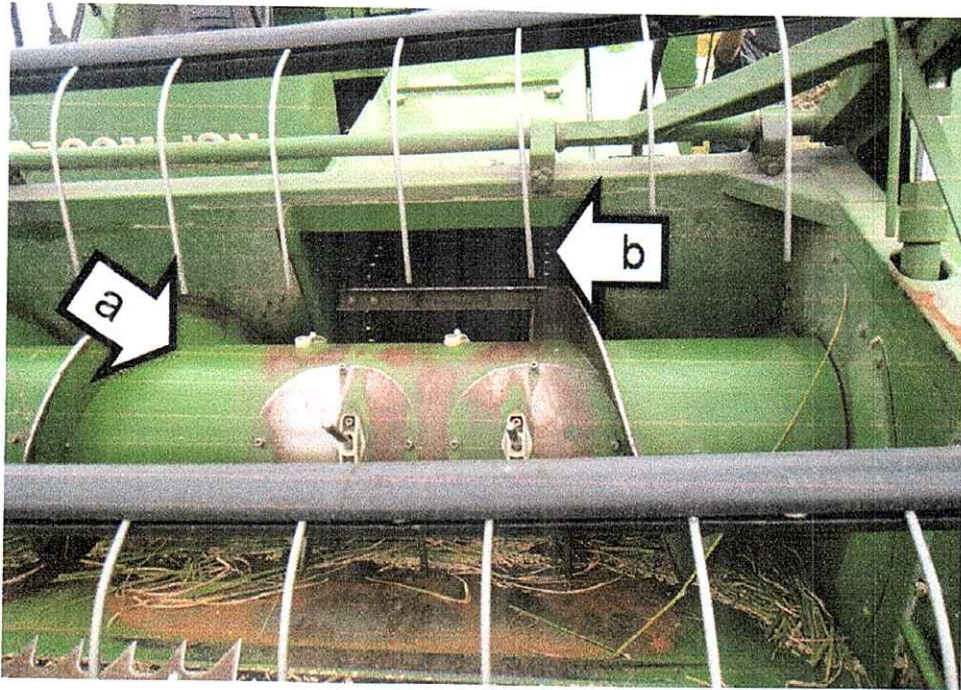


Figure 4. a) Screw conveyor
b) Chain conveyor

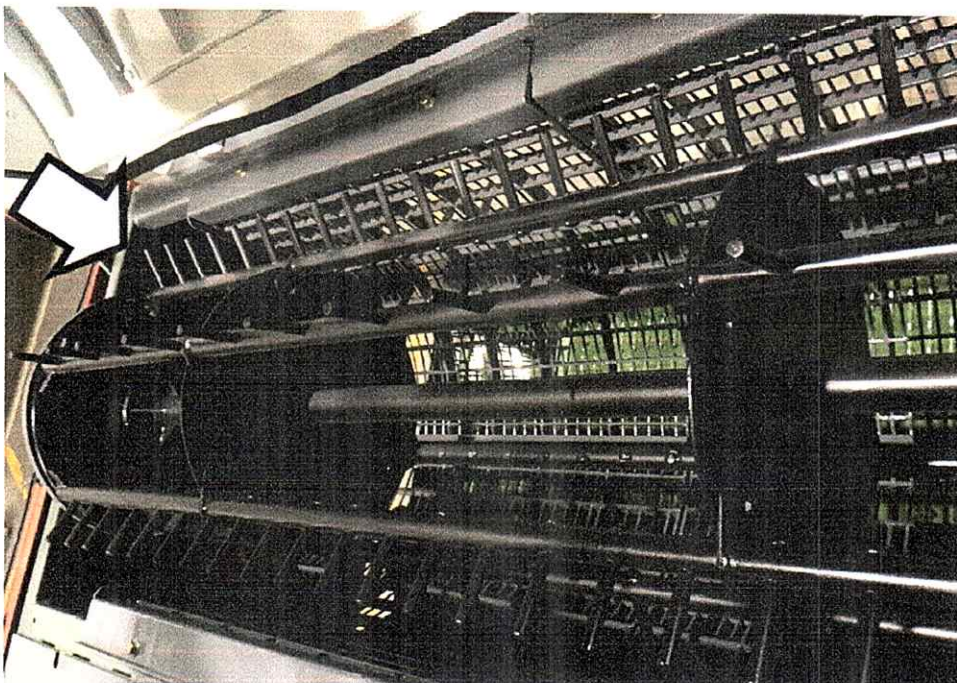


Figure 5. Threshing cylinders

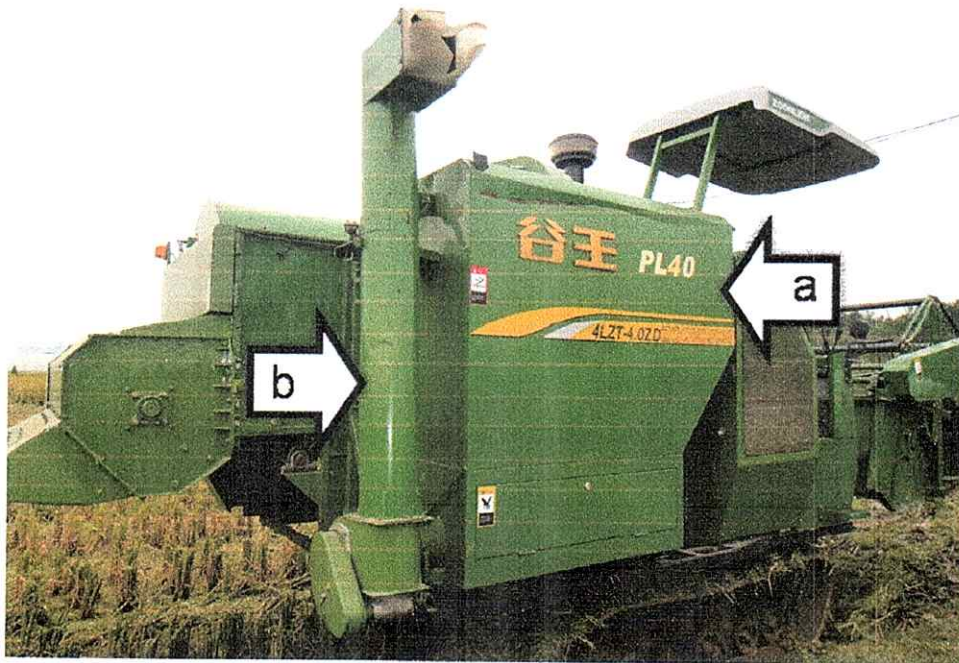


Figure 6. a) Grain Tank
b) Discharge chute



Figure 7. Control levers and control panel

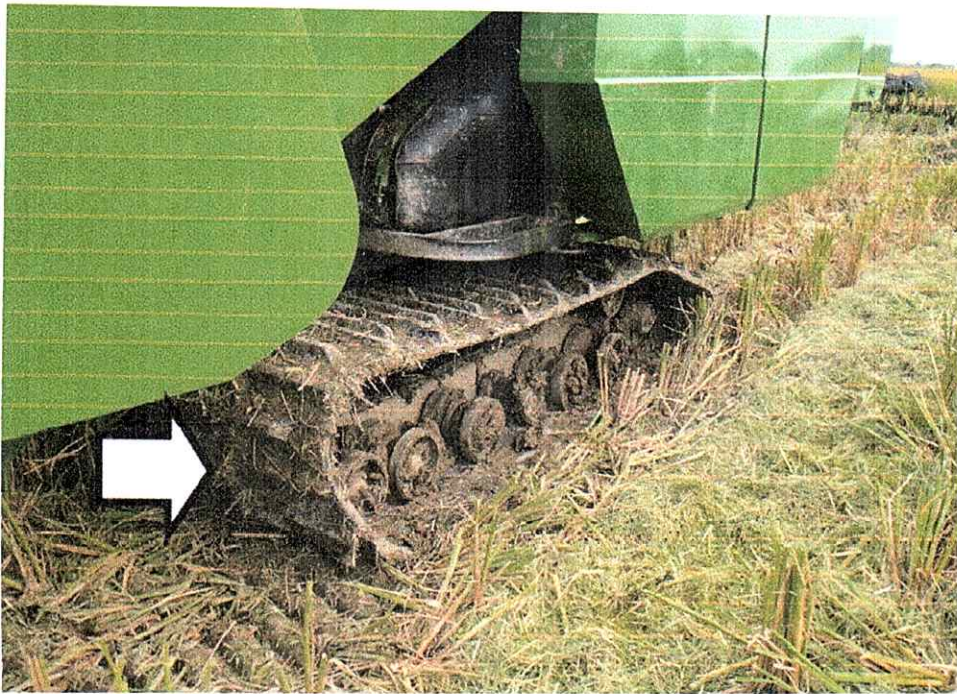


Figure 8. Crawler-type traveling device

RESULTS AND DISCUSSIONS

Results of field performance tests are shown in Table 1.

Performance

The performance of the rice combine was evaluated in terms of its field capacity, field efficiency, output capacity, grain losses and grain quality.

Actual Field Capacity. It is the ratio of the area harvested and the total time of operation. The Field Capacity of the machine was 0.378 ha/h or 2.65 h/ha.

Field Efficiency. It is the ratio of the Actual Field Capacity and the Theoretical Field Capacity, expressed in percent. Based on the Theoretical Field Capacity of 0.497 ha/h, the Field Efficiency of the machine was 76.1%.

Header Loss. This refers to the grains that have fallen to the ground due to machine's cutting operation. Header Loss was 0.11%.

Threshed Grain Output Capacity. The rice combine harvester had Threshed Grain Output Capacity of 4546 kg/h using "Hybrid" rice variety with a Grain Moisture Content of 20.7%, wet basis, Grain-Straw Ratio of 0.41 and average Plant Height of 970 mm.

Separation/Blower Loss. This refers to the grains that come out of the threshing chamber with the straw which is also the outlet of the blower. Separation/Blower Loss was 0.03 %.

Uncut Loss. This refers to the grains from the panicles that were not cut or missed by the reaping mechanism of the machine. There was no Uncut Loss.

Unthreshed Loss. This refers to grains that remained in the panicles of the plant fed into the threshing chamber. There was no Unthreshed Loss.

Grain quality. Quality of threshed grain was determined in terms of percent cracked and broken grains. Results showed that the threshed output had a net percentage Cracked Grains of 0.84% and 0.02% Broken Grain. (Cracked grains refer to those grains whose endosperms are partially fractured while broken grains refer to those grains that are split into 2 or more pieces.)

Fuel Consumption

The rice combine's 65.7 kW diesel engine had a Fuel Consumption Rate of 6.42 L/h.

Labor Requirement

During the test, four persons were needed to operate the rice combine: one for operating the machine, two for bagging and one for tying the sack of palay (Figure 9).

Observations

1. The rice combine was provided with canopy (**Figure 10a**) and an operator seat (**Figure 10b**) for the convenience of the operator.
2. When the grain tank was full, threshed grains were discharged through the folding-type discharge chute (**Figure 11**) with screw conveyor.
3. The transmission systems of the combine were provided with covers.
4. The rice combine was easy to maneuver at corners (**Figure 12**).
5. The rice combine was provided with stickers (**Figure 13**) on safety with symbols and writings in English language.



Figure 9. Operators of the rice combine



Figure 10. a) Canopy
b) Operator seat



Figure 11. Discharge chute for threshed grains



Figure 12. Turning at corner

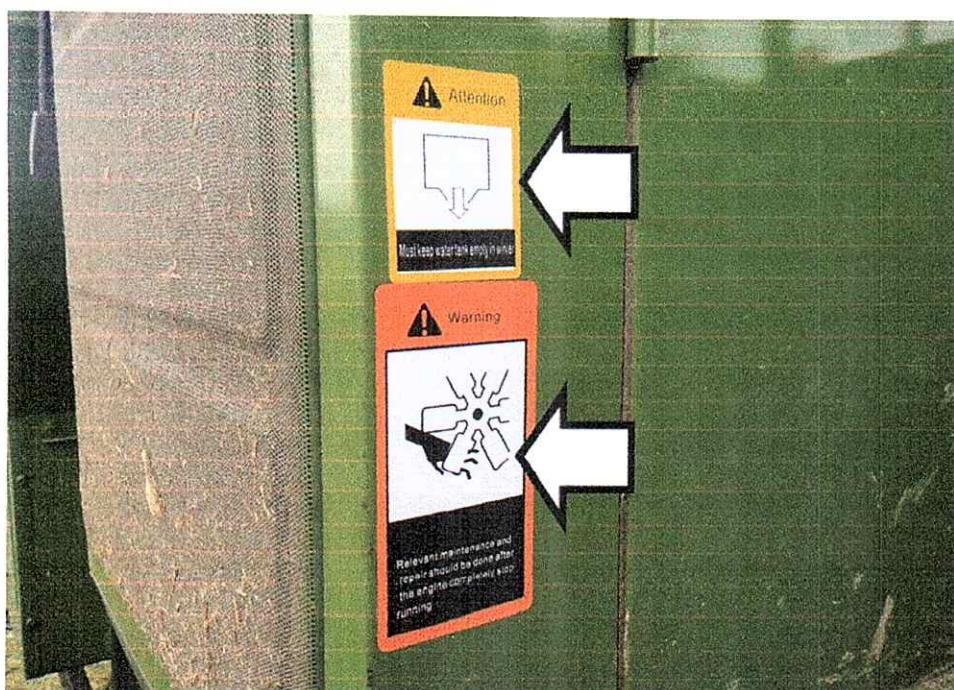


Figure 13. Stickers attached to the rice combine

FIELD PERFORMANCE TEST

Machine tested : ZOOMLION Rice Combine Harvester
Model : 4LZT-4.0ZD

Location of test: Nanling County
Wuhu City, China

Date of test : October 10, 2015

Table 1. Results of field performance test

ITEMS		TEST DATA
A.	Condition of Crop	
1.	Variety	: Hybrid
2.	Mode of Planting	: Transplanted
3.	Spacing of Plants, row x hill (mm)	: 300 x 200
4.	Plant Height (mm)	: 970
5.	Circumference of Stem at Cutting Height (mm)	: 95
6.	Grain-Straw Ratio	: 0.41
7.	Angle of Stem Inclination, degrees	: 90
8.	Moisture Content of Grains (%)	: 20.7
B.	Field Conditions	
1.	Type of Soil	: Clay loam

Table 1. Results of field performance test (cont'n.)

ITEMS		TEST DATA
2.	Soil Resistance (kg/cm ²) :	4.90
C. Field Performance		
1.	Method of Operation :	Circuitous
2.	Travelling Speed (kph) :	4.25
3.	Effective Cutting Width (mm) :	2000
4.	Height of Cut or Stubbles (mm) :	355
5.	Actual Field Capacity	
	a) ha/h :	0.378
	b) h/ha :	2.65
6.	Theoretical Field Capacity	
	a) ha/h :	0.497
	b) h/ha :	2.01
7.	Field efficiency (%) :	76.1
8.	Threshed Grain Output Capacity (kg/h) :	4546
9.	Threshed grain unloading rate (kg/h) :	7721
10.	Purity of threshed grain (%) :	97.31
11.	Header Loss (%) :	0.11
12.	Separation/Blower Loss (%) :	0.03

Table 1. Results of field performance test (cont'n.)

ITEMS		TEST DATA
13.	Unthreshed Loss (%)	: Nil
14.	Uncut Loss (%)	: Nil
15.	Total Grain Loss (%)	: 0.14
16.	Noise level, db(A)	: 95.2
17.	Fuel consumption, (L/h)	: 6.42
D. Laboratory Grain Analysis		
1.	Before Harvest	
	a) Cracked (%)	: 0
	b) Broken (%)	: 0
2.	After Harvest	
	a) Cracked (%)	: 0.84
	b) Broken (%)	: 0.02
3.	Net Percent	
	a) Cracked	: 0.84
	b) Broken	: 0.02

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MAR 30 2016
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N.B.

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